# Predicting the Culturability of Marine Bacterial Genomes Using Greedy and Monte Carlo Algorithmic Approaches

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## 1 Introduction

The majority of bacteria present in natural environments, including marine ecosystems, are considered unculturable, meaning they cannot be grown in laboratory conditions using conventional techniques [5]. While some estimates suggest that only 1% of environmental bacteria are culturable, recent advances in microbial cultivation methods have challenged this paradigm, increasing the proportion of bacteria with closely related cultured representatives [3]. Nevertheless, many marine bacterial strains remain unculturable, limiting our ability to study their physiology, ecological roles, and potential biotechnological applications. This challenge is particularly significant in marine microbiology, where culturing remains essential for biodiscovery, such as identifying novel natural products with pharmaceutical potential. The inability to culture these bacteria restricts access to their metabolic capabilities, highlighting the need for predictive methods to assess their culturability.

To address this challenge, we propose an algorithmic approach to determine the culturability of marine bacterial genomes based on genomic signatures. Specifically, we aim to identify distinguishing sequence patterns, or k-mers, that are significantly associated with culturable bacterial genomes. By leveraging existing datasets of culturable and unculturable marine bacterial genomes, we develop and compare brute-force and greedy algorithms to extract characteristic k-mer sets that differentiate culturable strains from unculturable ones. Our work seeks to provide a systematic and computationally efficient means of predicting bacterial culturability, which could guide future efforts in microbial isolation and cultivation. Moreover, by optimizing our algorithm to handle full-length bacterial genomes or 16S rRNA sequences, we aim to refine our predictive framework and contribute to a deeper understanding of marine microbial ecology.

## 2 Problem Formulation

In this study, we aim to identify genomic signatures associated with the culturability of marine bacterial genomes. Let us consider the following mathematical definitions:

**A** and **B** are two different string sets, where $A$ represents culturable bacterial genomes and $B$ represents unculturable bacterial genomes.

**d_match(v, S, d)**: Given a string $v$, a string set $S$, and a non-negative integer $d$, we say that $v$ *d-matches* $S$ if there exists a substring of $v$ that approximately matches one of the strings in $S$ with at most $d$ mismatches: (i.e. Hamming distance less than $d$)

$$d_match(v,S,d) \iff \exists s \in S, \exists i \text{ such that } H(v[i:i+k],s) \leq d \tag{1}$$

**coverage(S, V, d)**: The coverage of $S$ over a set $V$ is defined as the fraction of strings $v \in V$ that d-match $S$. The coverage function measures how well a given set $S$ of k-mers covers a string set $V$:

$$\text{coverage}(S,V,d) = \frac{|\{v \in V \mid d_match(v,S,d)\}|}{|V|} \tag{2}$$

where $d_match(v,S,d)$ is true if there exists a substring of $v$ that approximately matches at least one string in $S$ with at most $d$ mismatches.

**score(A, B, S, d)**: Given two sets of strings $A$ and $B$, and a set of k-mers $S$, we define the score function as:

$$\text{score}(A,B,S,d) = \text{coverage}(S,A,d) - \text{coverage}(S,B,d). \tag{3}$$

The score function quantifies the difference in coverage between the culturable and unculturable genome sets, where a higher score indicates that the selected k-mers in $S$ are more representative of culturable genomes than unculturable ones.

**Input:** String sets $A$ (culturable genomes) and $B$ (unculturable genomes). A positive integer $k$ representing k-mer length. A non-negative integer $d$ representing mismatch tolerance.
**Output:** A string set $S$ maximizing $\text{score}(A,B,S,d)$ among all possible sets of k-mers:

$$S^* = \arg\max_{S \subseteq \Sigma^k} \text{score}(A,B,S,d), \tag{4}$$

## 3 Algorithmic Challenge

Our goal is to identify a representative set of k-mers $S$ that maximizes the culturability score function $\text{score}(A,B,S,d)$, where $A$ represents culturable bacterial genomes and $B$ represents unculturable bacterial genomes. This optimization problem involves searching for an optimal subset $S$ from a large set of candidate k-mers extracted from $A$. Given that the total number of possible k-mer subsets is exponential in size, this problem falls into the category of NP-hard combinatorial optimization problems, meaning no known polynomial-time solution exists.
A brute-force approach could be used to obtain the optimal solution by enumerating all possible subsets of k-mers and selecting the subset that maximizes $\text{score}(A,B,S,d)$. The number of possible subsets can be determined by considering that we must choose any subset from a pool of $|C|$ candidate k-mers, where $|C|$ is the total number of distinct k-mers extracted from $A$:

$$\sum_{i=0}^{|C|} \binom{|C|}{i} = 2^{|C|} \tag{5}$$

This exponential growth in the number of subsets makes the brute-force approach infeasible for large datasets. Even for moderate values of $|C|$, the number of subsets quickly becomes computationally intractable. Therefore, an optimized approach is necessary. Given the impracticality of finding a globally optimal solution, we adopt a greedy approach that balances computational efficiency with solution quality, potentially finding locally optimal solutions that approximate the global optimum.

### 3.1 Brute-Force Algorithm and Runtime Analysis

The brute-force approach explores all possible k-mer subsets to find the optimal set $S$. The pseudocode is shown in Algorithm 1.

#### 3.1.1 Runtime Analysis

**Step 1: Extracting k-mers from** $A$ Let $N$ be the number of sequences in $A$ and $L$ be the length of each sequence. Each sequence contributes approximately $(L-k+1)$ k-mers. In the worst case (if all k-mers are unique), the total number of k-mers is:

$$|C| = N \times (L-k+1) \tag{6}$$

**Step 2: Brute-force search over the power set of** $C$ The power set of $C$ contains $2^{|C|}$ candidate subsets. For each candidate subset, we compute the score function, which requires evaluating the coverage function over $A$ and $B$. This operation has a complexity of:

$$O((N+M) \times |C| \times L \times k) \tag{7}$$

where $M$ is the number of sequences in $B$.
**Overall Complexity:** Since we iterate over all $2^{|C|}$ subsets and evaluate the scoring function for each, the total runtime complexity of the brute-force approach is:

**Algorithm 1** Brute Force Characteristic K-mer Set

**Input** Sets of sequences $A$ and $B$, range for k-mer length $(k_{min}, k_{max})$, mismatch threshold $d$
**Output** Selected k-mer set $S$

```
C ← ∅                                                   ▷ Initialize an empty set for k-mers
for each sequence in A do
    for i ← 0 to length(sequence) − k do
        kmer ← substring(sequence, i, k)
        Add kmer and its variants to C
    end for
end for
bestScore ← −∞
bestSet ← ∅                                             ▷ Best candidate k-mer set
for each candidateSet S in PowerSet(C) do
    currentScore ← score(A, B, S, d)
    if currentScore > bestScore then
        bestScore ← currentScore
        bestSet ← S
    end if
end for
return bestSet
```

$$O(2^{|C|} \times (N+M) \times |C| \times L \times k) \tag{8}$$

**Conclusion:** Due to the exponential number of candidate subsets, this brute-force approach is computationally infeasible for large-scale bacterial genome datasets. As a result, we require a more efficient optimization approach, such as a greedy algorithm combined with Monte Carlo sampling, which will be discussed in the next section.

# 4 Our Proposed Algorithm

## 4.1 Count Dictionary Construction

The count dictionary is a data structure that maps each unique k-mer to the set of indices of sequences in which it appears. The psuedo code is shown in Algorithm 2.

**Algorithm 2** Construct Count Dictionary

**Input** Set of sequences $G$, range for k-mer length $(k_{min}, k_{max})$, mismatch threshold $d$
**Output** Dictionary *count* mapping k-mers to sequence indices

```
Initialize count ← empty dictionary
for each sequence g_i ∈ G do
    for k = min_k to max_k do
        for each k-mer kmer in g_i do
            for all variant v of kmeer with at most d mismatches do
                Add i to count[v]
            end for
        end for
    end for
end for
return count
```

## 4.2 Greedy Algorithm

The greedy approach repeatedly adds the k-mer with the highest gain, which is defined as the genome counts this k-mer can cover in set $A$ minusing the genome counts covered in set $B$, to the selected k-mer set $S^*$. The gain is calculated with the count

dictionary to speed up and the process stops when none of the non-selected k-mers give positive gain. The pseudo code of this approach is shown in Algorithm 3.

**Algorithm 3** Greedy Algorithm

**Input** Count dictionaries *A_cover* and *B_cover*
**Output** Selected k-mer set $S^*$

```
Initialize A_cover ← ∅, B_cover ← ∅ and S* ← ∅
while True do
    Initialize gain* ← −inf
    for each k-mer kmer in A_count do
        gain ← (A_count[kmer] − A_cover) − (B_count[kmer] − B_cover)
        if gain is greater than gain* and kmer is not in S* then
            gain* ← gain, kmer* ← kmer
        end if
    end for
    if gain* > 0 then
        Add kmer* to set S* and Update A_cover & B_cover
    else
        break
    end if
end while
return S*
```

## 4.3 Monte Carlo Algorithm [4]

This method introduces randomness by selecting k-mers at random and stopping when no improvement is observed for a predefined threshold of iterations. The procedure will be run several times and output the set $S$ giving the best score. The pseudo code of this approach is shown in Algorithm 4.

**Algorithm 4** Monte Carlo Algorithm

**Input** Count dictionaries *A_cover* and *B_cover*, negative counter threshold *neg_cnt_thres*, number of trials $T$
**Output** Selected k-mer set $S^*$

```
Initialize S* ← ∅ and best score score* ← −inf
for t = 1 to T do
    Initialize A_cover ← ∅, B_cover ← ∅, S ← ∅, and neg_cnt ← 0
    while neg_cnt is less than neg_cnt_thres do
        Randomly sample a k-mer kmer from A_count
        gain ← (A_count[kmer] − A_cover) − (B_count[kmer] − B_cover)
        if gain is greater than 0 then
            Add kmer to set S* and Update A_cover & B_cover
            neg_cnt ← 0
        else
            Add 1 to neg_cnt
            if neg_cnt = neg_cnt_thres then
                break
            end if
        end if
    end while
    if score of S is greater than score* then
        S* ← S
        score* ← score of S
    end if
end for
return S*
```

## 4.4 Modified Greedy Algorithm

Finally, we proposed a modified greedy algorithm that incorporates randomness to mitigate the risk of getting stuck in a locally optimal solution. This approach runs the greedy algorithm within a Monte Carlo framework, introducing stochasticity when searching for k-mer with the highest gain. Specifically, when a k-mer with a higher gain is found during the search, a random value between 0 and 1 is sampled. The k-mer is only selected as the new optimal choice if the sampled value exceeds a predefined threshold. This modification enhances exploration and improves the likelihood of finding a better overall solution. The pseudo code of this approach is shown in Algorithm 5.

**Algorithm 5** Modified Greedy Algorithm

**Input** Count dictionaries *A_cover* and *B_cover*, threshold $\theta$, number of trials $T$
**Output** Selected k-mer set $S^*$

```
Initialize S* ← ∅ and best score score* ← −inf
for t = 1 to T do
    Initialize A_cover ← ∅, B_cover ← ∅ and S ← ∅
    while True do
        gain* ← −inf
        for each k-mer kmer in A_count do
            gain ← (A_count[kmer] − A_cover) − (B_count[kmer] − B_cover)
            if gain is greater than gain* and kmer is not in S* then
                if (a random value from 0-1) > θ then
                    gain* ← gain
                    kmer* ← kmer
                end if
            end if
        end for
        if gain* > 0 then
            Add kmer* to set S* and Update A_cover & B_cover
        else
            break
        end if
    end while
    if score of S is greater than score* then
        S* ← S
        score* ← score of S
    end if
end for
return S*
```

# 5 Experimental Design

## 5.1 Dataset

**Set A**, our representative culturable sequences, were made up of 16S rRNA sequences from a 2020 study titled Diversity and distribution of marine heterotrophic bacteria from a large culture collection, from Isabel Sanz-Sáez et al [1]. This study sought to characterize global abundance of heterotrophic marine bacteria. Data was accessed from GenBank with accession numbers MH731309-MH732621 and MK658870-MK659428.

**Set B**, our representative unculturable sequences, were accessed from NCBI, under the nucleotide database, with the search conditions “(16S[All Fields] AND rRNA[All Fields]) AND "uncultured marine bacterium"[porgn] AND (bacteria[filter] AND biomol_rrna[PROP])”.

Had we had the opportunity to extend our work to full genomes, we would have drawn Set A from a 2021 study, Phylogenetic distribution of polysaccharide-degrading enzymes in marine bacteria [6] . This work investigated 961 polysaccharide-degrading enzymes from 2,182 high-quality marine bacteria genomes, with accession numbers available in the paper’s Supplementary Table 3. We would have drawn Set B from the 2022 work The OceanDNA MAG catalog contains over 50,000 prokaryotic genomes

originated from various marine environments [2]. Through their work to develop a new genome reconstruction pipeline, they reconstructed genomes of 8,466 species (52,325 total genomes), accessible under project number PRJDB11811 at the European Nucleotide Archive. Due to these genomes coming from metagenomic sampling efforts, we would have proceeded with the assumption that the majority of these genomes were from unculturable bacteria.

## 5.2 Computational Complexity Discussion

The first step in all three algorithms is the construction of the *count dictionary*, which records the index of each k-mer and its variances across the sequence sets $A$ and $B$. Given $N$ as the total number of sequences with length $L$ and k-mers ranging from $k_{\min}$ to $k_{\max}$, the complexity of constructing the count dictionary is $O(NLk_{\max})$. Since all algorithms rely on this dictionary for k-mer selection, this step represents a shared computational cost.

The subsequent computational complexity varies across the three algorithms based on their selection strategies and the inclusion of randomness. The **Greedy Algorithm** iterates through all k-mers in the count dictionary *A_count*, selecting the one with the largest gain in each iteration. The complexity of this selection process is $O(|C|)$, leading to a total complexity of $O(NLk_{\max} + |C|)$. However, the greedy nature of the algorithm means it only considers locally optimal choices at each step. The algorithm is logically correct, but there are limitations because the greedy approach does not explore the entire global search space.

To address this limitation, the **Monte Carlo Algorithm** introduces randomness by performing multiple independent runs to explore different k-mer selections. This increases the complexity to $O(TNk_{\max}|C|)$, where $T$ represents the number of Monte Carlo iterations. This added computational cost provides a more diverse search space, potentially leading to better global solutions at the expense of increased runtime.

Finally, the **Modified Greedy Algorithm** strikes a balance between efficiency and exploration. It retains the greedy strategy of selecting high-gain k-mers but incorporates randomness by accepting a k-mer only if a randomly sampled value exceeds a set threshold. This adjustment allows for greater flexibility in k-mer selection while maintaining computational efficiency. This modification increases the complexity to $O(NLk_{max} + T|C|)$, which offers a more efficient approach to the Monte Carlo while improving performance over the original greedy method.

Figure 1 presents the score distribution from 300 trials of the **Monte Carlo Algorithm** and the **Modified Greedy Algorithm** on the full dataset. The k-mer sets obtained using **Modified Greedy Algorithm** consistently achieved higher scores compared to those from the **Modified Greedy Algorithm**. Although both algorithms share the same computational complexity, the **Modified Greedy Algorithm** reached a score greater than 0.75 in just five trials, whereas the **Monte Carlo Algorithm**, even after 300 trials, struggled to reach a score of 0.60.

These results highlight the advantage of incorporating guided randomness into the greedy approach, striking a balance between exploration and exploitation. The **Modified Greedy Algorithm** not only maintains efficiency but also significantly improves performance in identifying optimal k-mer sets.

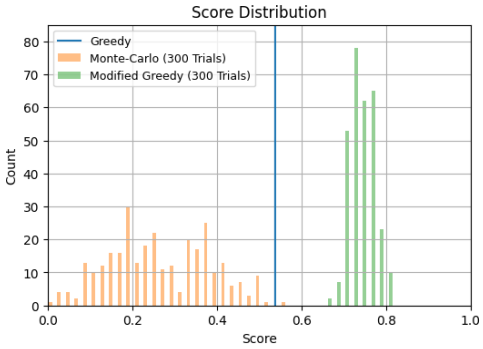



Figure 1: Score Distribution

## 5.3 Experimental Result

The results in Table 1 highlight a clear trend in algorithm performance across different conditions. The Greedy algorithm consistently runs the fastest but tends to achieve lower scores, as it is likely to be stuck in a local optimum, as expected. Monte Carlo, while improving scores with more trials, comes at a significant computational cost, often requiring much longer runtimes. Modified Greedy consistently outperforms both, striking the best balance between accuracy and efficiency. It achieves the highest scores across all conditions while maintaining reasonable runtimes, making it the most effective approach overall. These results suggest that while Monte Carlo can enhance performance with sufficient trials, Modified Greedy is the most optimal choice when balancing runtime and accuracy.

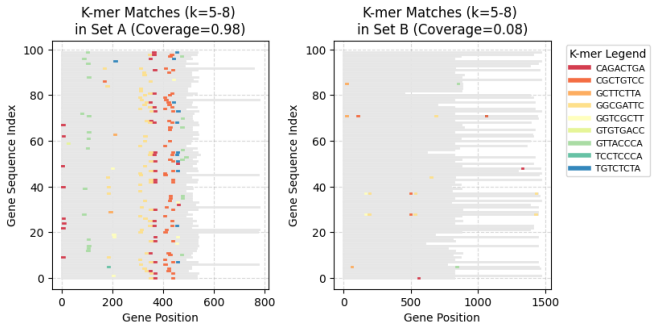
Our experimental results demonstrate the effectiveness of our greedy + Monte Carlo algorithm in identifying genomic segments associated with culturability across marine bacterial genomes. Using a dataset of 1900+ bacterial genome sequences, our algorithm successfully identified a set of characteristic k-mers that appear consistently across all culturable genomes while being significantly less frequent in unculturable genomes. This provides strong evidence that our algorithm can isolate genetic features that are indicative of a bacterium's ability to be cultured.

| Sets A and B | k_range | d | Algorithm | # of Trials | # of kmers | Runtime | Score |
|---|---|---|---|---|---|---|---|
| All 16S | 5-8 | 1 | Greedy | – | 8 | 0:18 | 0.538 |
| | | | Monte Carlo | 300 | 14 | 3:56 | 0.570 |
| | | | Modified Greedy | 20 | 17 | 2:11 | 0.810 |
| 100 each 16S | 5-8 | 1 | Greedy | – | 8 | 0:03 | 0.80 |
| | | | Monte Carlo | 100 | 7 | 1:01 | 0.65 |
| | | | Modified Greedy | 20 | 9 | 0:10 | 0.90 |
| 100 each 16S | 9-14 | 1 | Greedy | – | 8 | 0:46 | 0.97 |
| | | | Monte Carlo | 50 | 6 | 29:20 | 0.89 |
| | | | Modified Greedy | 20 | 8 | 2:32 | 1.0 |
| 100 each 16S | 15-18 | 1 | Greedy | – | 8 | 0:45 | 0.830 |
| | | | Monte Carlo | 20 | 21 | 19:07 | 0.940 |
| | | | Modified Greedy | 50 | 10 | 09:41 | 0.990 |

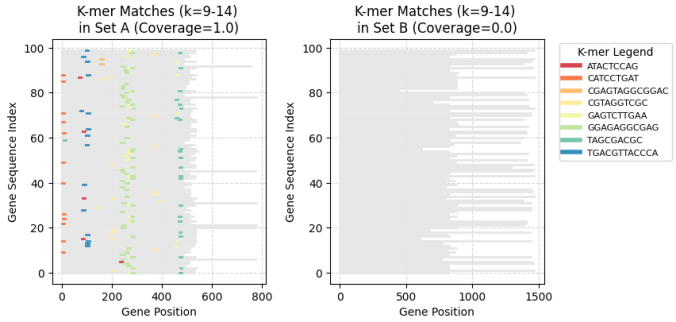
Table 1: Results of different algorithms under various conditions

Figure 3a shows the distribution of the identified k-mer sequences across the genome sequences in both the culturable set $A$ and the unculturable set $B$. A striking observation from these results is that in the culturable genome set, these k-mers appear at similar positions across different genomes, suggesting that our algorithm has correctly identified conserved genetic elements related to culturability. This is an important discovery, as it suggests that a computational approach can be used to pinpoint genetic markers that are most representative of a given biological trait, in this case, culturability.

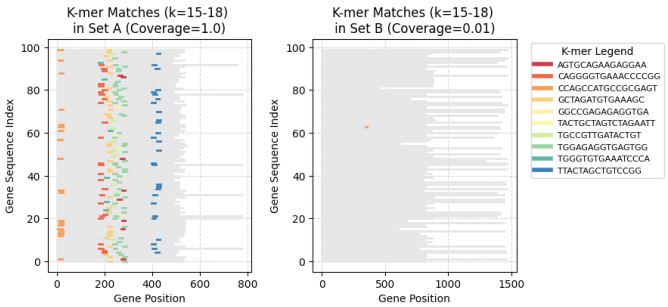
Additionally, the algorithm proved to be computationally efficient. Compared to the brute-force approach, which is infeasible for large datasets due to its exponential complexity, our greedy + Monte Carlo method identified characteristic k-mers in a fraction of the time while maintaining accuracy. This efficiency makes our approach suitable for large-scale genomic analyses, providing a practical tool for predicting the culturability of bacteria from metagenomic datasets.




(a) K-mer Matches (k=5-8)

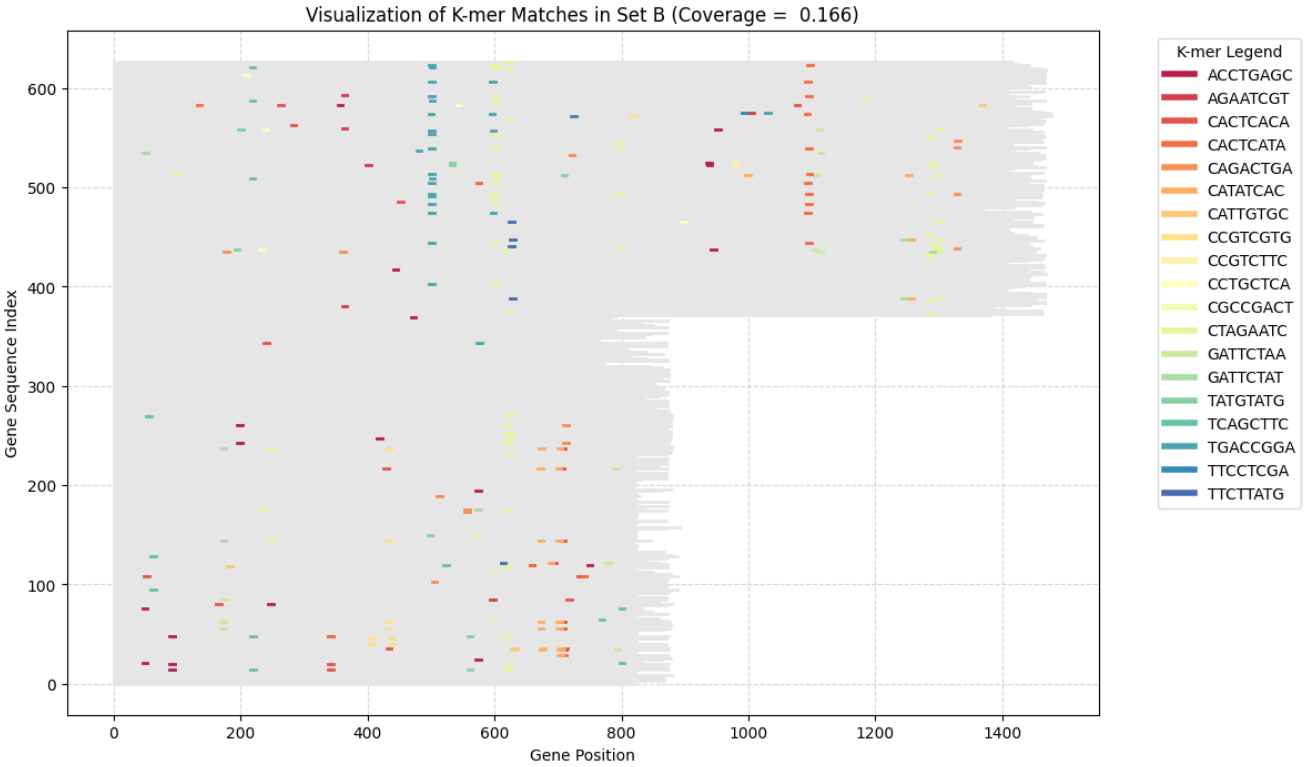



(b) K-mer Matches (k=9-14)

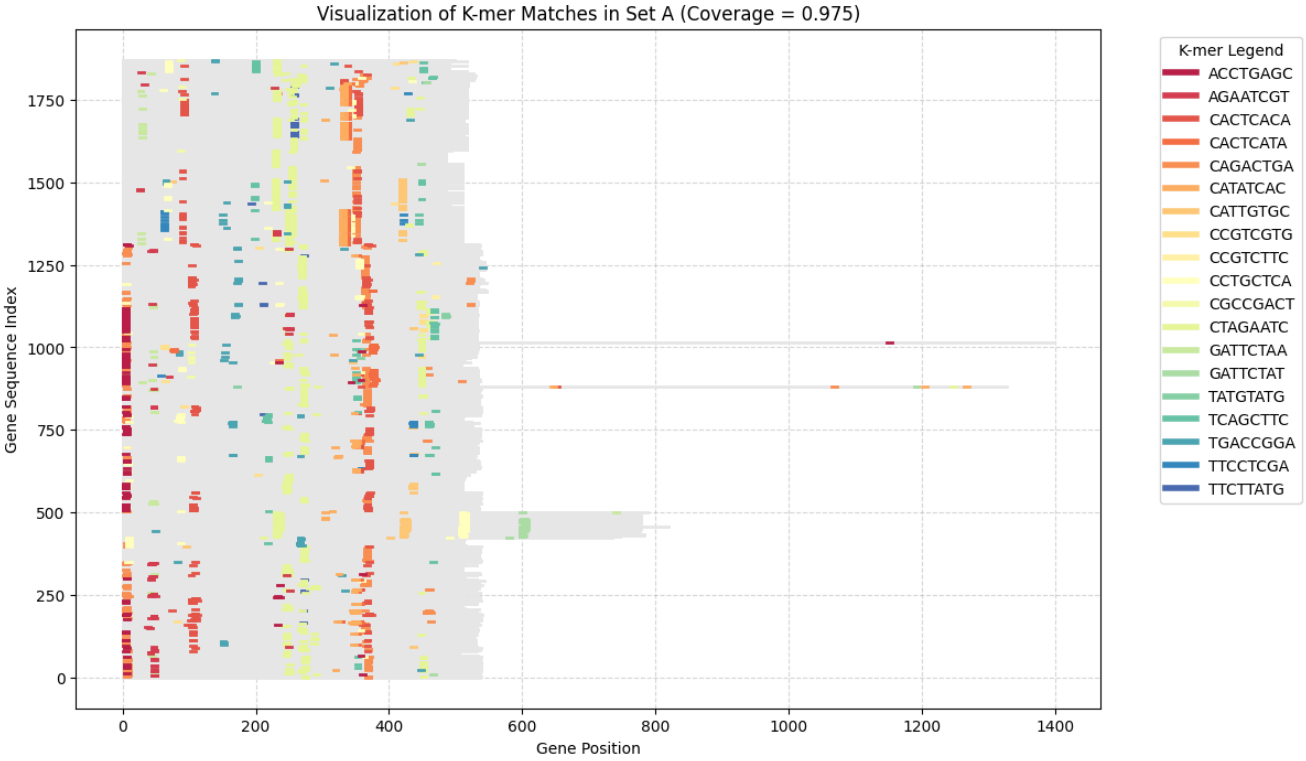



(c) K-mer Matches (k=15-18)

Figure 2: Visualization of k-mer matches across 100 randomly-picked 16S genome set under different k-mer length range.




(a) Full Set B Coverage




(b) Full Set A Coverage

Figure 3: Visualization of k-mer matches across the entire 16S genome set.

## 6 Conclusion

We have developed a modified greedy algorithm that is capable of identifying a set of k-mers of varying lengths that are highly abundant in a set of 16S rRNA genes of culturable marine bacteria, and less abundant in a set of 16S rRNA genes from uncultured marine bacteria. While we were limited by computational memory and time from exploring whole genomes that fall into these respective categories, our success on this smaller scale is highly promising. Given our output k-mers from our modified greedy algorithm, we would be able to search an arbitrary 16S rRNA sequence from an unknown marine bacteria, and offer a reasonable estimate of if this bacteria could be cultured.

If we could have extended our work into the whole-genome datasets described above, it is likely that we would have identified regions of the genome that could serve as biomarkers for potential ability to be cultured in a conventional laboratory setting.

## References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